

Enterprise Cloud Infrastructure & DevSecOps Patterns

Vision 2026 Reference Architecture for Microsoft Azure

```
resource azurerm_kubernetes_cluster 'aks_cluster' {  
  name           = 'main-aks-cluster'  
  location       = azurerm_resource_group.main.location  
  resource_group_name = azurerm_resource_group.main.name  
  ...  
}
```

```
resource azurerm_application_gateway 'app_gw' {  
  name           = 'main-app-gateway'  
  location       = azurerm_resource_group.main.location  
  resource_group_name = azurerm_resource_group.main.name  
  sku {  
    name = 'Standard_v3'  
    tier  = 'Standard_v2'  
    ...  
  }  
}
```



THE FOUNDATIONAL ENGINEERING PRINCIPLES



ZERO-TRUST IDENTITY

Compound Auth & OIDC

Enforces Workload Identity Federation and eliminates static service principal secrets.



IaC SOVEREIGNTY

100% Terraform Standard

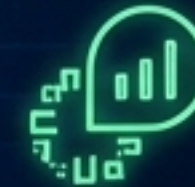
Declarative state enforcement. Azure Portal modifications are classified as configuration drift and automatically overwritten.



IMMUTABLE INFRASTRUCTURE

Stateless Design

Prioritizes resource replacement over patching. Complete environment reproducibility from code.



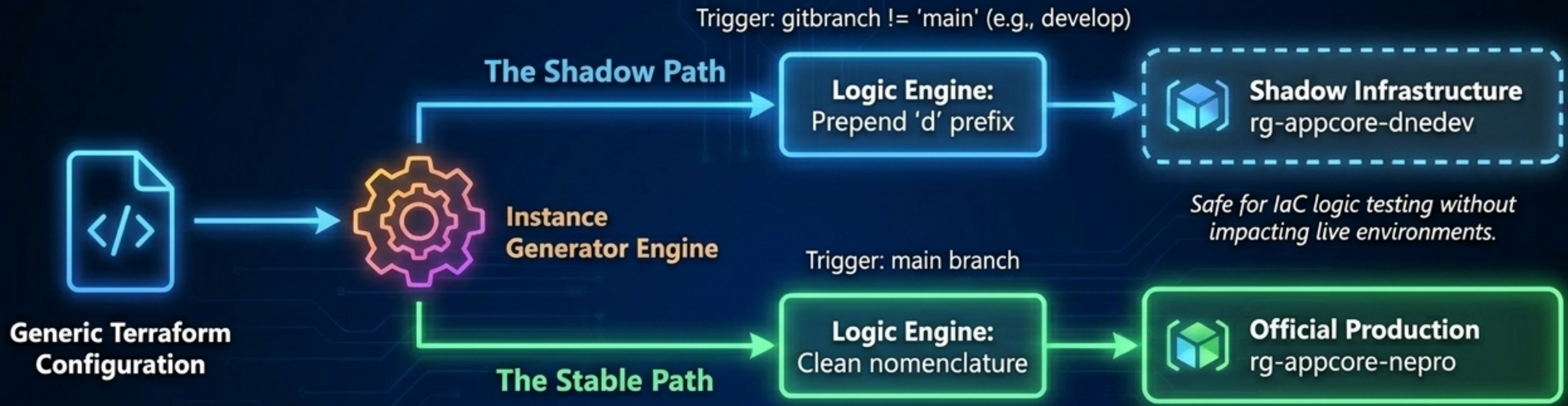
OBSERVABILITY-BY-DESIGN

Native Telemetry

No resource exists without immediate integration into regional Log Analytics Workspaces.

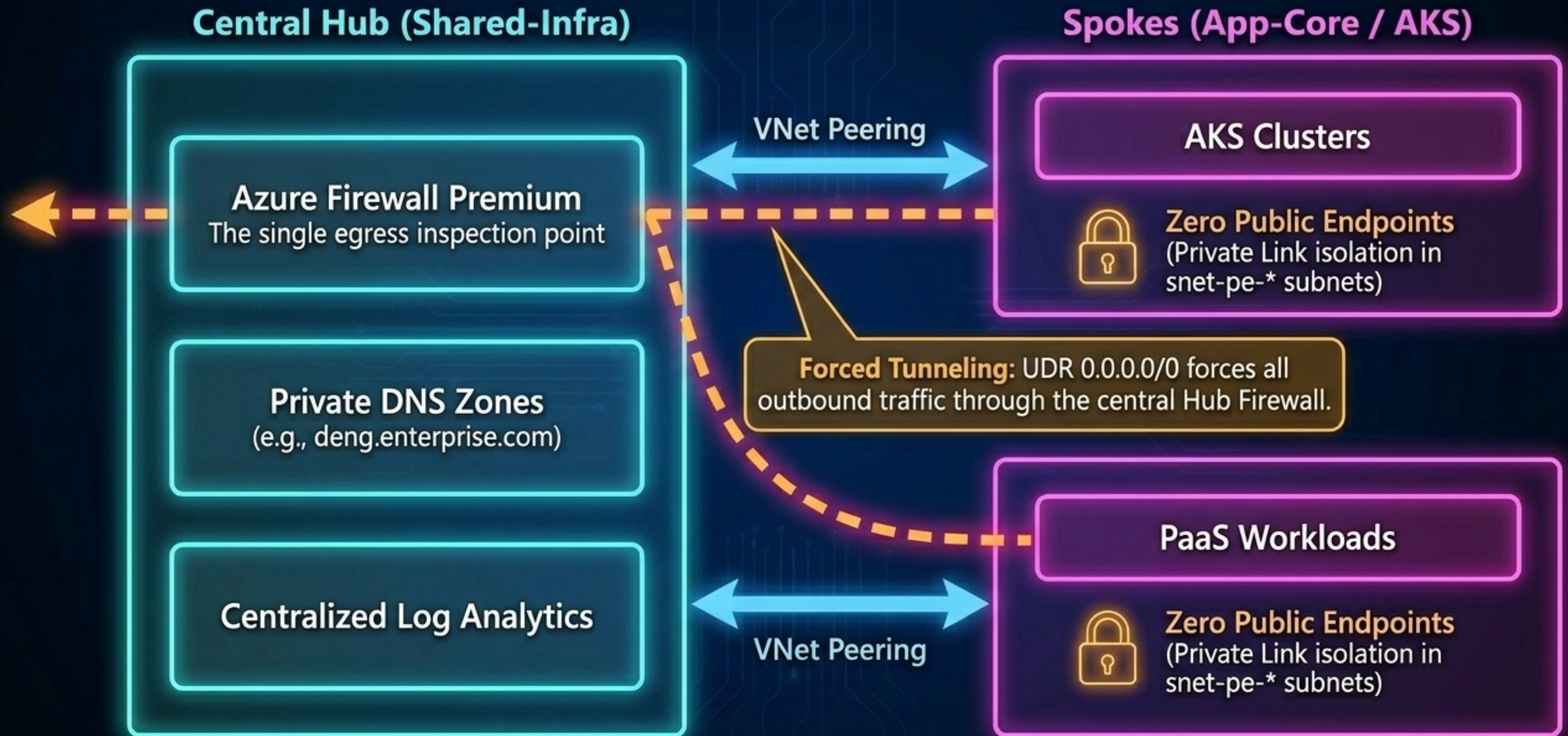


THE DRY LOGIC ENGINE: BRANCH-AWARE ENVIRONMENT GENERATION

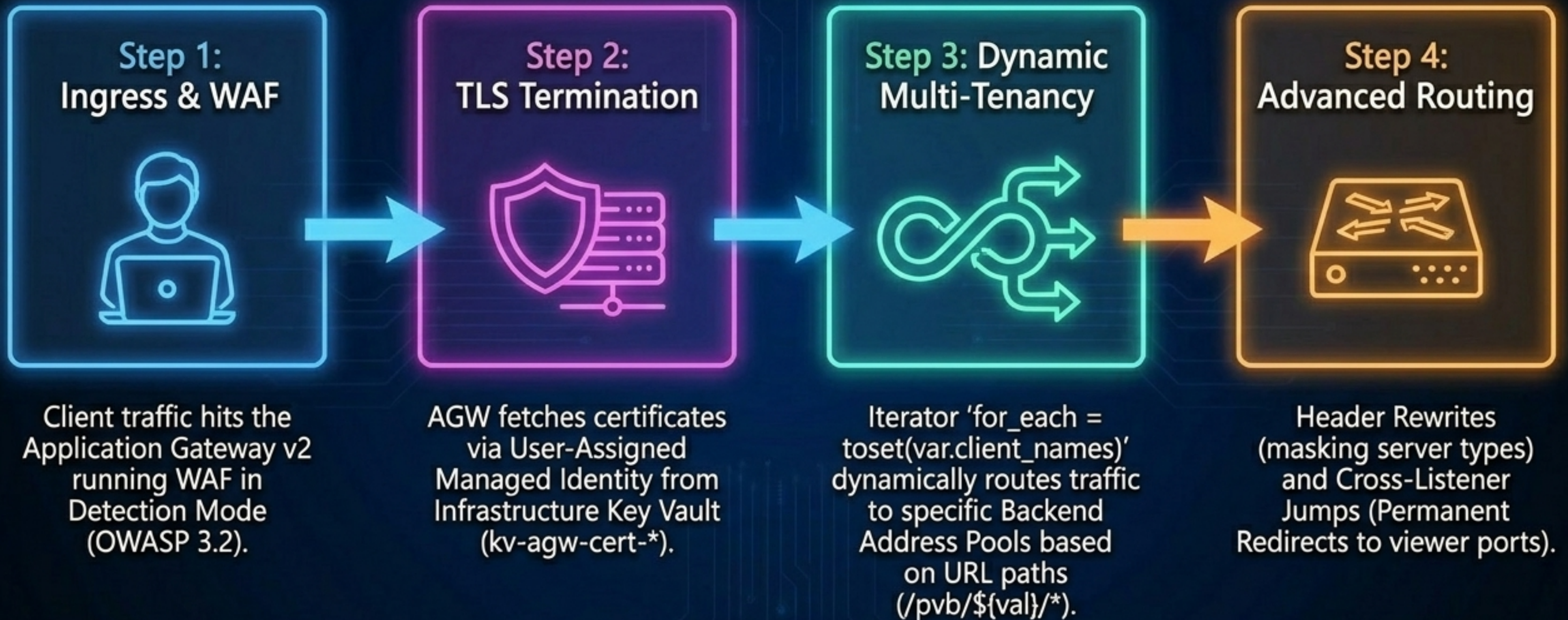


Naming Prefix	Collision Risk	Target Audience
'd' (e.g., dnedeve)	Zero (Isolated by naming)	Platform Engineers / SRE
None (e.g., nepro)	High (Requires PR approval)	App Developers / End Users

THE SYMMETRICAL HUB-SPOKE NETWORK BACKBONE

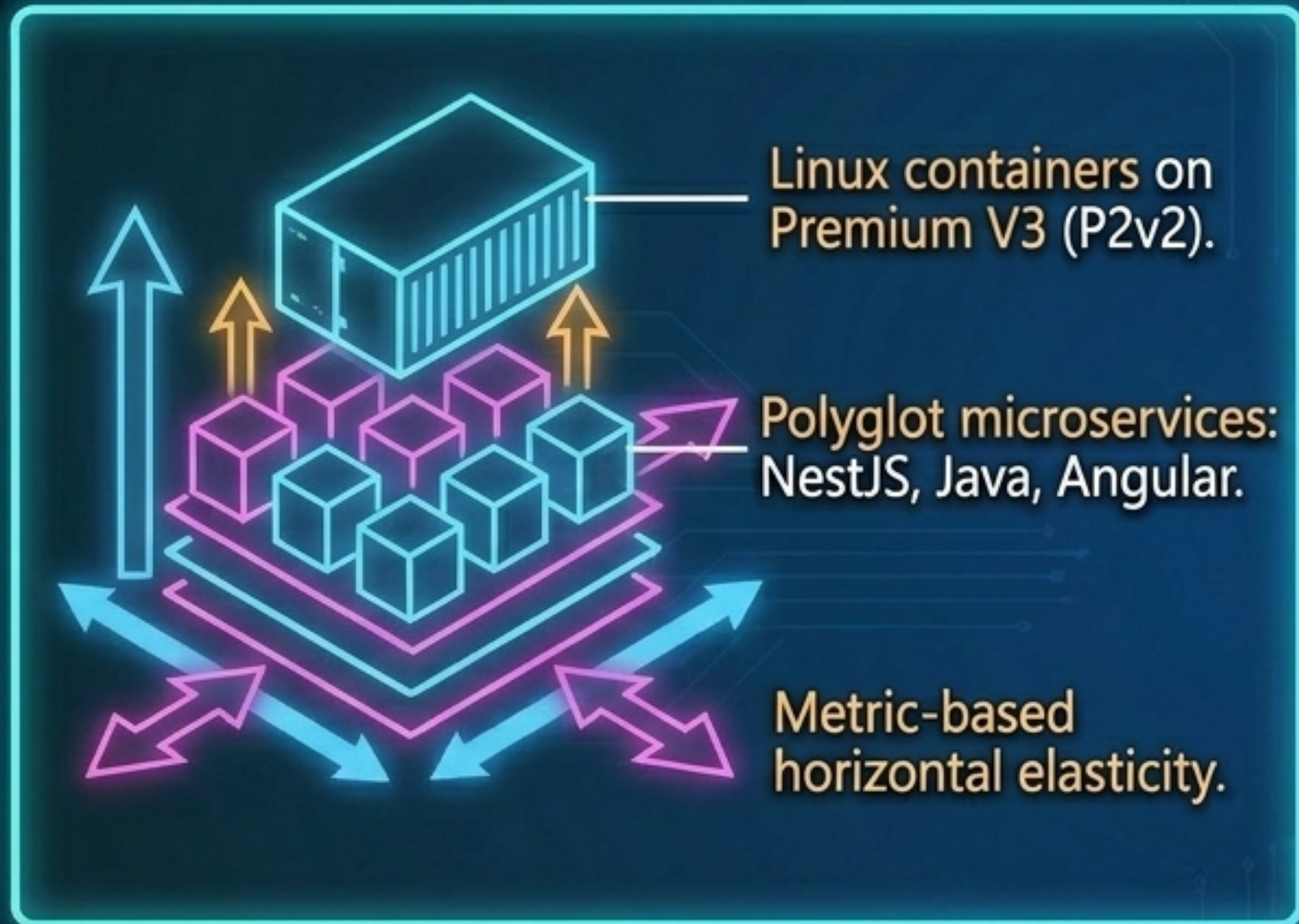


EDGE SECURITY & L7 TRAFFIC ORCHESTRATION



Compute Hub: Polyglot PaaS & AKS Machine Learning

PaaS Ecosystem (App Services)



AKS & ML Inference

System Pool (appspool001)

Standard_DS2_v2 for core infra (Ingress, DNS).

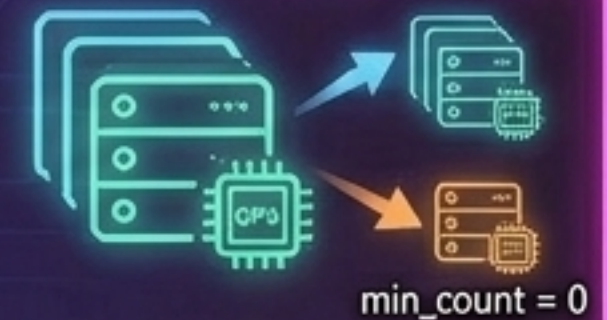


User Pool (appspool002)

Standard_E8as_v4 for ML High-Memory/GPU.

Label isolation: app = mljobs

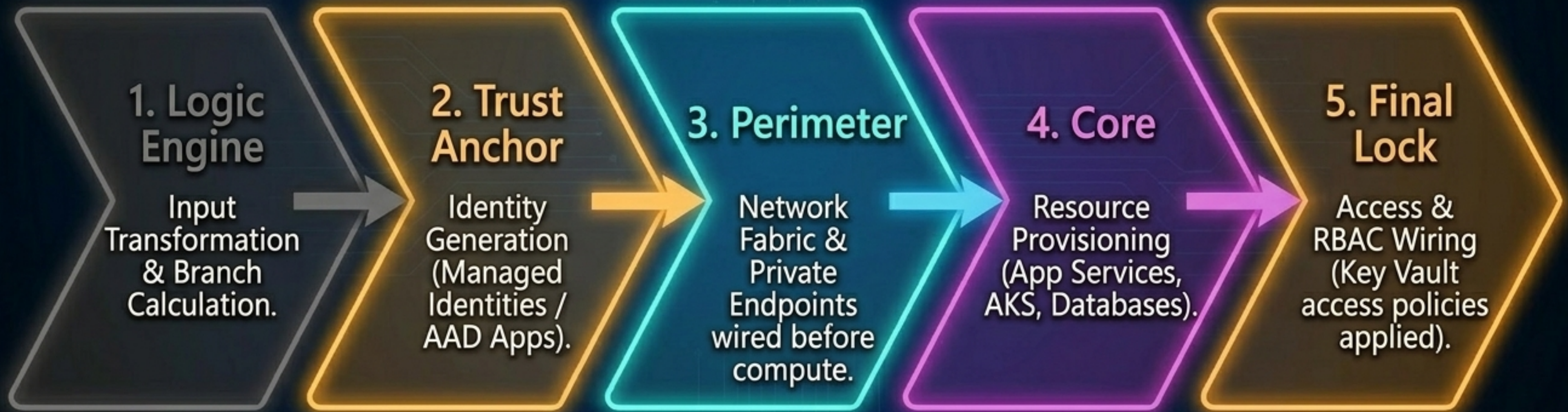
Cost optimization: Scale-to-Zero (min_count = 0).



Network Fabric: Azure CNI with Dynamic Pod IP Allocation.
Dedicated /19 subnets prevent node IP exhaustion.

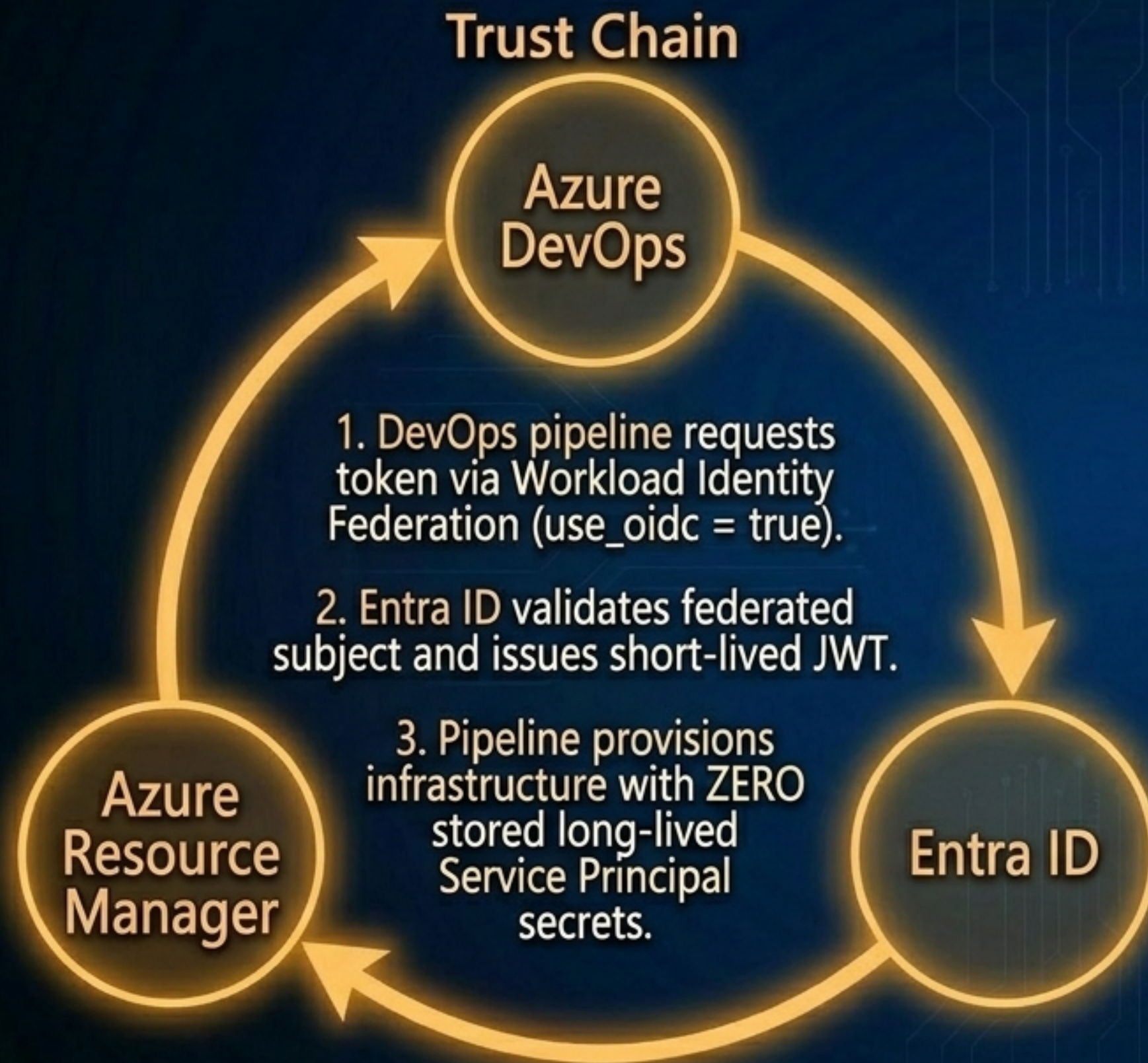
Domain-Driven Composite Module Architecture

Atomic (AVM)	Composite (This Repo)
Manual wiring	Pre-wired dependencies
High maintenance	Secure by default



A **single module call** inherently wires security and networking before any compute is deployed.

Zero-Trust Pipeline: Secretless CI/CD Orchestration



Global Pipeline Dependency Flow



Governance-First: The Compound Identity Model



The OAuth 2.0 On-Behalf-Of (OBO) Flow



If either key is missing, access is mathematically blocked (403 Forbidden).

Database Strategy & Data Sovereignty Decision Matrix

	Scenario A: MongoDB Atlas Standard	Scenario B: MongoDB Atlas Hardened	Scenario C: Azure Cosmos DB (Vision 2026 Target)
Security	IP Allowlisting (Legacy)	Entra ID SSO	100% Native Azure RBAC (No local users)
Operational Complexity	Low	High (Requires cross-cloud federation)	Lowest (Native HCL)
Networking	Public internet traverse (Avoid)	Private Link hourly fees	VNet Sovereign (Zero egress fees)

Conclusion: Identity is the new perimeter. Cosmos DB transforms the persistence layer from an external dependency into a sovereign Azure component.

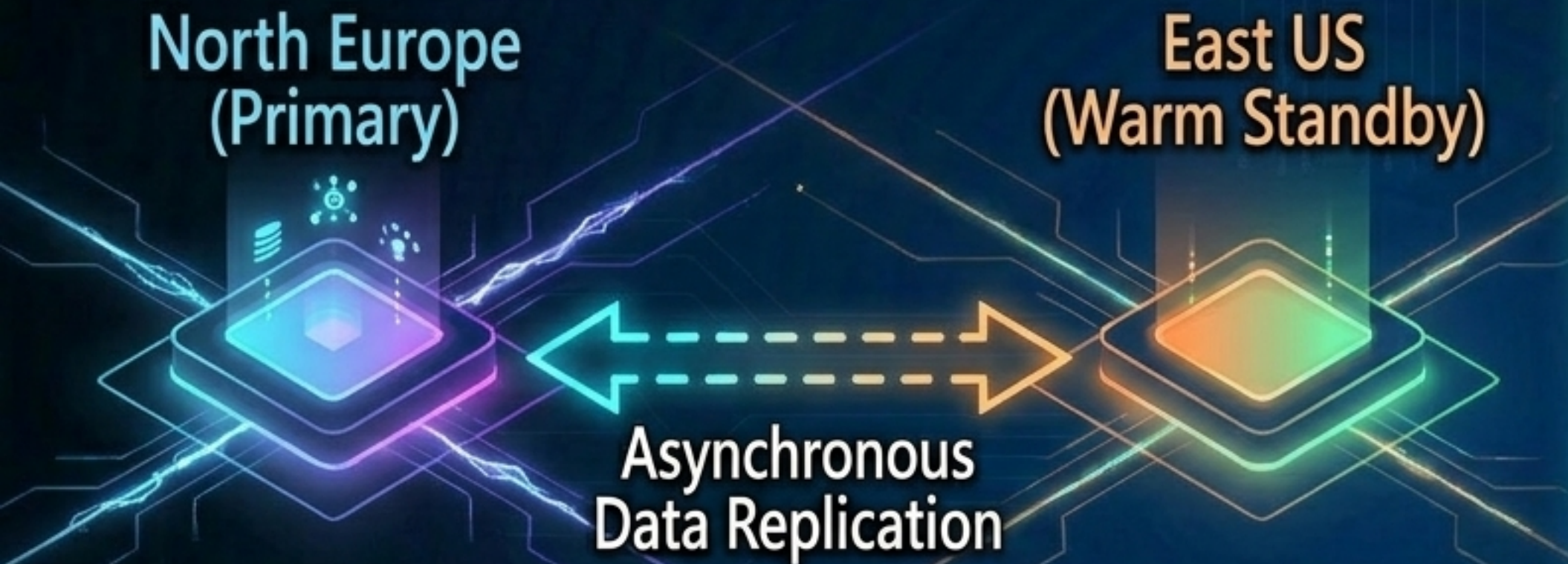
Scaling Intelligence: The 4-Tier Elasticity Model



GreenOps Impact

Implementation of **"scale_down_utilization_threshold = 0.5"** and **AKS Scale-to-Zero** for ML nodes actively reduces the carbon footprint during idle hours.

Multi-Region **Warm Standby** & IaC Rehydration



- Atlas/Cosmos **Global Zones** (Active-Active)
- **Immutable Backups** in Recovery Services Vault (180+ days retention, Point-in-Time)

The 'Zero-to-Hero' Rehydration Sequence

- 1 Force-Unlock **.tfstate** blobs to clear pipeline locks.
- 2 Rebuild Network Foundation (Shared-Infra).
- 3 Instantiate Compute Core (AKS/ACR).
- 4 Deploy Application Logic & Restore Data.

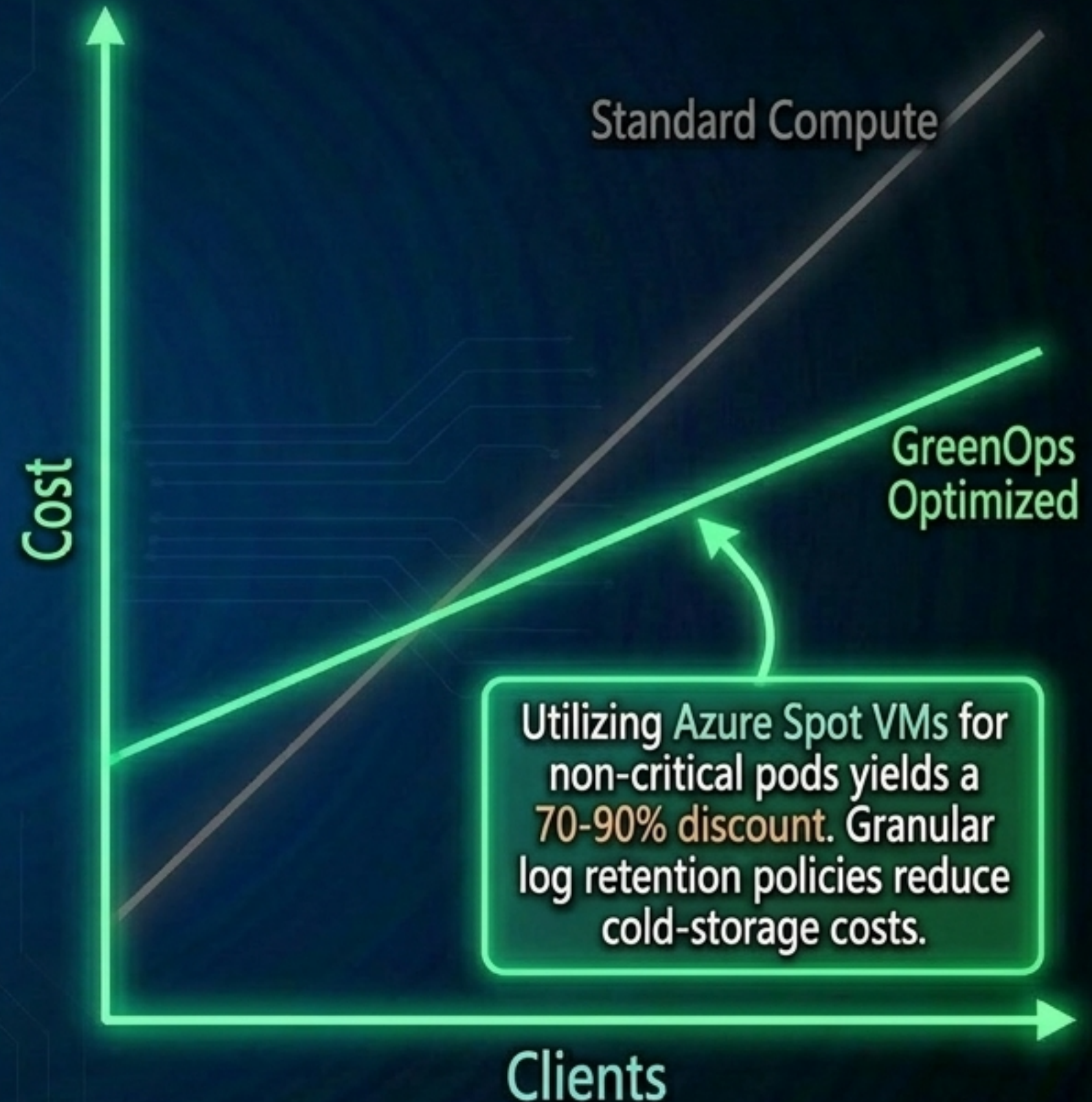
FinOps Architecture & Cost Optimization Engineering

The Cost Baseline



The 'Shadow Infrastructure' Multiplier

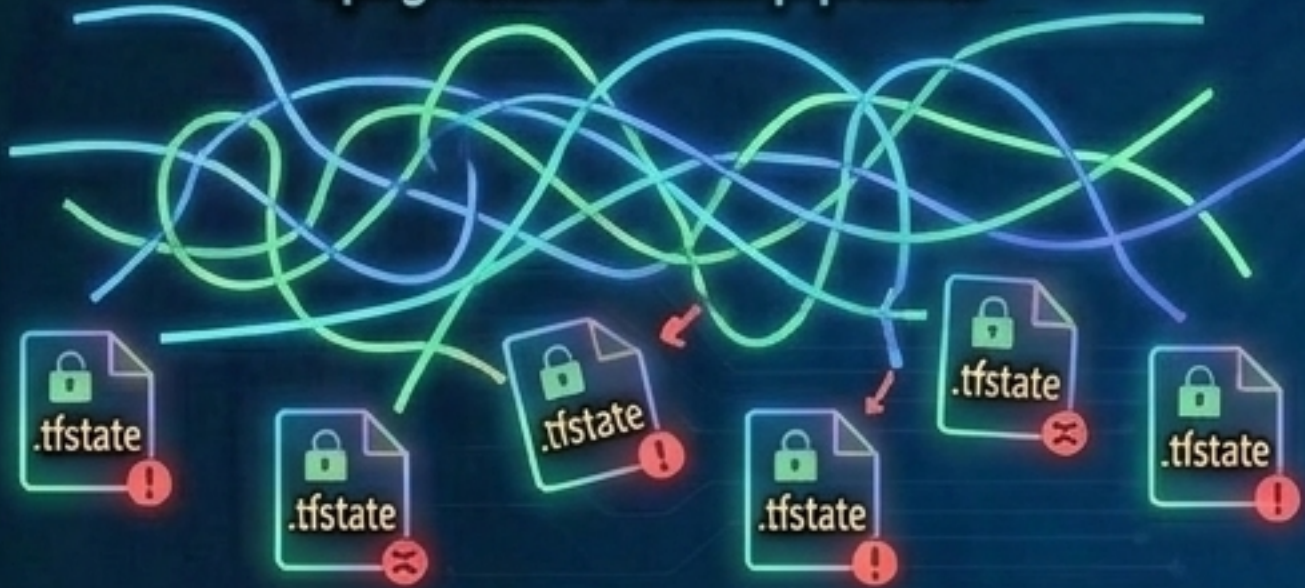
Testing branches (develop) intentionally duplicate the entire stack, generating a **2x cost multiplier** during active IaC engineering phases.



Vision 2026: AI-Ready Infrastructure & Native Terraform Stacks

Legacy (Pipeline-Driven Orchestration)

Spaghettable YAML pipelines



Fragmented state files, manual dependency management, and sequential execution bottlenecks.

Vision 2026 (Native Terraform Stacks)

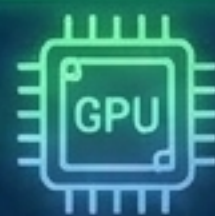


A single unified HCL component model. Automatic parallelization, cross-layer planning, and unified dependency graphs.

The AI-Ready Perimeter



Zero-Trust Identity
(OIDC/OBO)



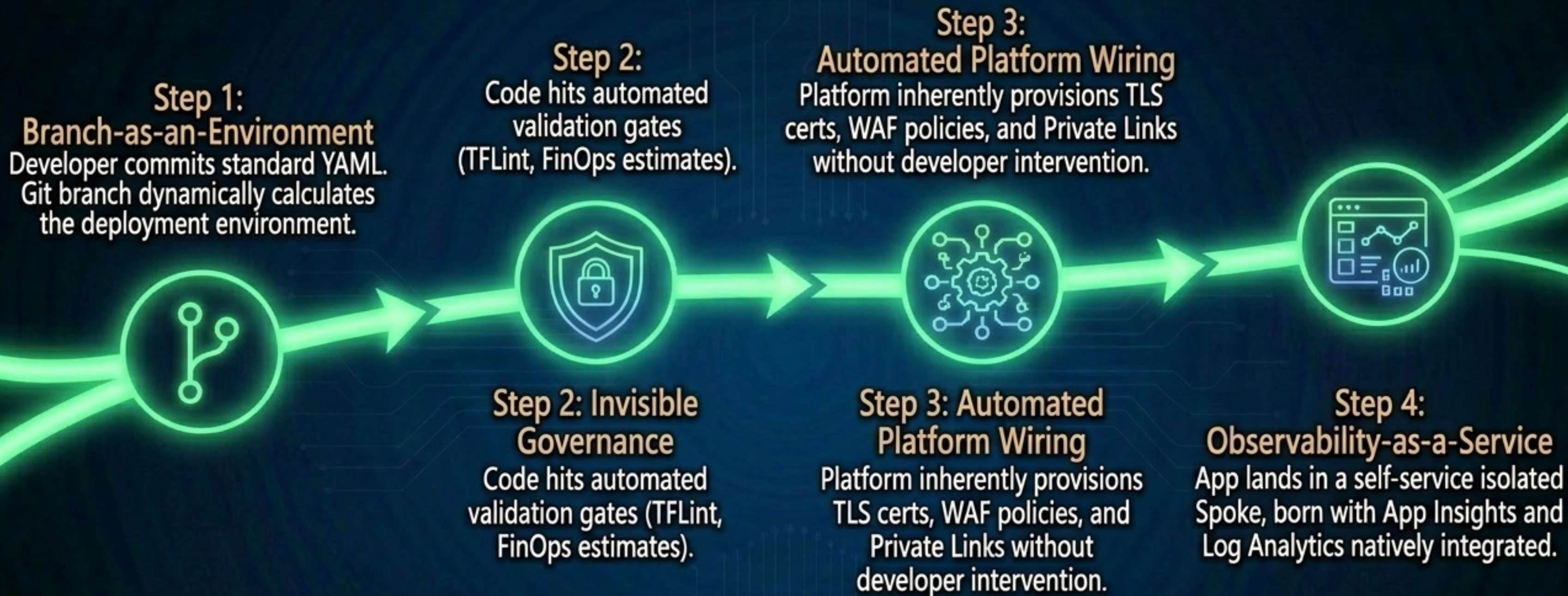
Sovereign ML Compute
(GPU scale-to-zero)



Vector Database Readiness
(Cosmos VNet Sovereign)

Primed for autonomous AI Agents.

The Internal Developer Platform (IDP) Golden Path



Takeaway: Developers consume standards; they do not reinvent pipelines.
Maximum DevEx through Infrastructure-as-a-Product.